

A03



Given: $P = 400 \text{ LB}$

$E = 1000$

$$\delta = \frac{PL}{AE}$$

Find: $L = ?$ $A = ?$ $\delta = 0.009 \text{ in}$

$$L = \frac{\delta AE}{P}$$

$D = 1 \text{ in}$ $A = 0.25 \text{ in}^2$

$$L = \frac{(0.009 \text{ in})(0.25 \text{ in}^2)(1000 \frac{\text{LB}}{\text{in}^2})}{400 \text{ LB}}$$

$$L = 176.7 \text{ in}$$